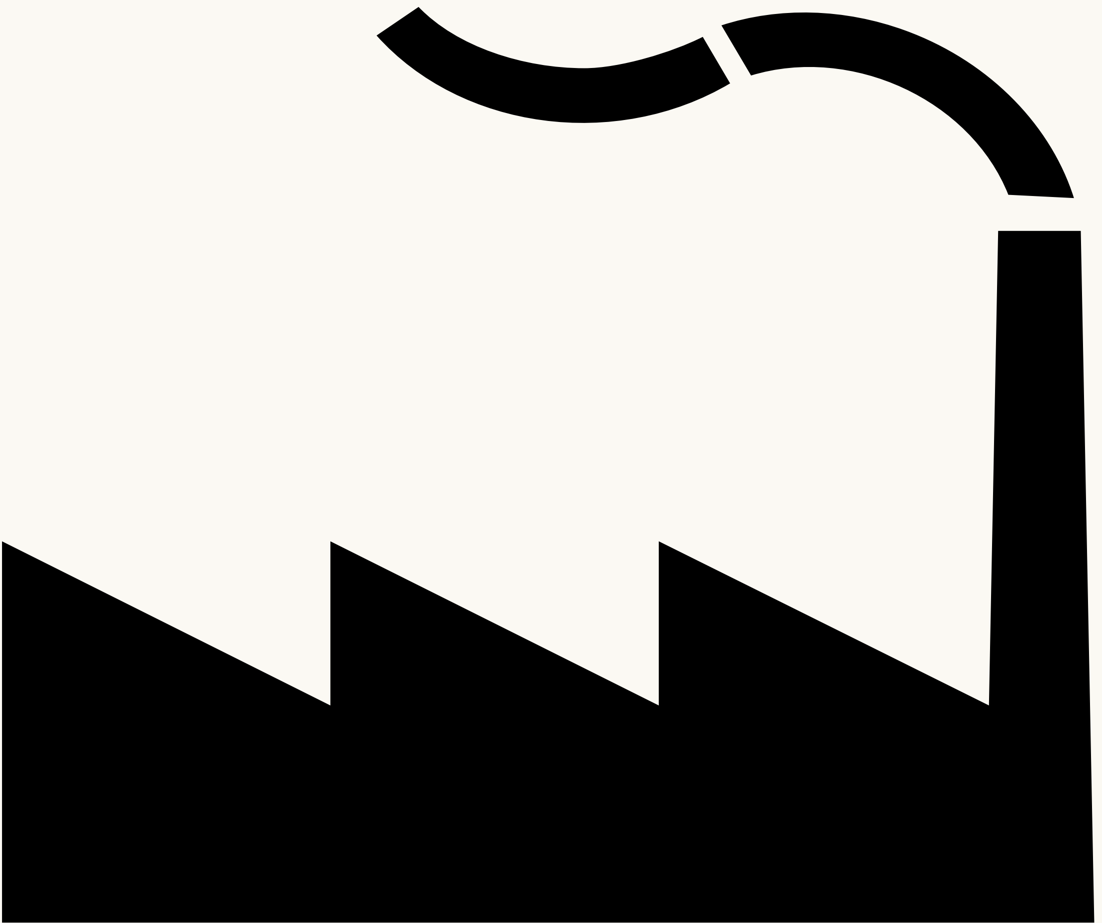
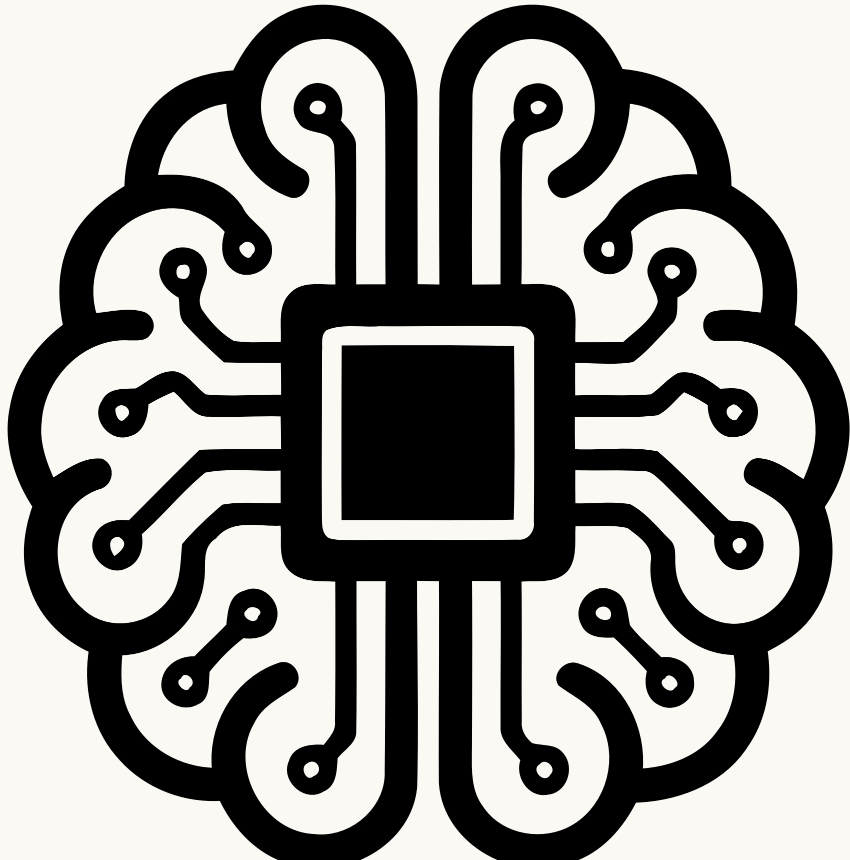
predictive maintenance

Machine Learning for Industrial Machine Failure Prediction

Predicting failures before they happen. Prevent downtime. Reduce losses



- **Industrial AI Project**
- **Designed by : Enas Khaled**



The Problem :

focus especially with small business owners

Unexpected Machine Failures :

→ machines can fail without warning.

Production Downtime :

→ Unexpected failures stop production.

Financial Losses :

→ Downtime leads to lost production and higher costs.

Emergency Maintenance :

→ Technicians are often called after the failure happens.

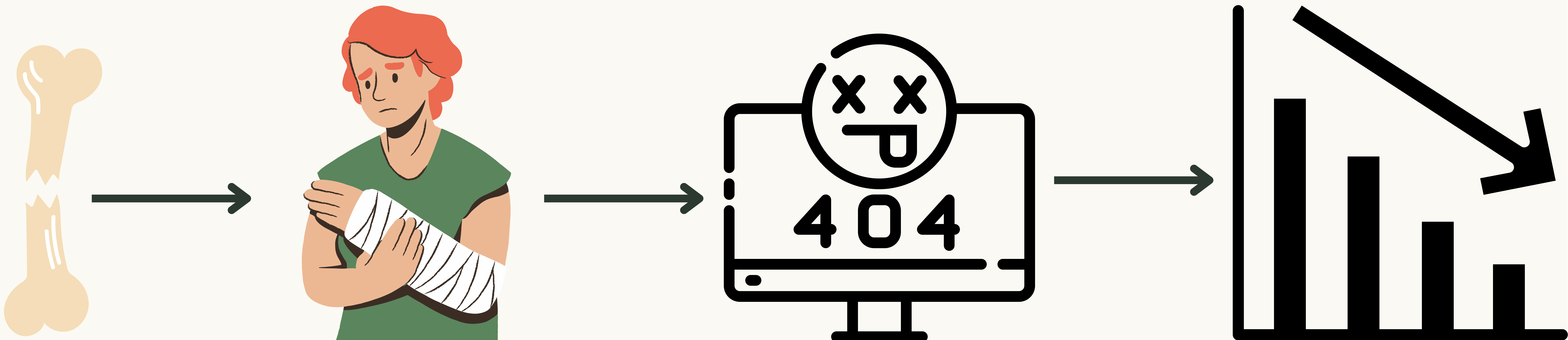
Machine Failure → ***Production Stops***

“Factories don't just lose machines

they lose time, production, and money”

Financial Loss ← ***Downtime***

A broken bone can disrupt the rest of the body.



How Machine Learning Helps?

Analyze Machine Data:

It analyzes historical machine data and identifies patterns related to failures.

Predict Failures

The model uses these patterns to predict the possibility of a future machine failure.

Early Alerts

When a potential failure is detected, the system can alert the maintenance team.

Preventive Action

Technicians can inspect and maintain the machine before an unexpected breakdown

Industrial AI Project

Factories machines problems

Designed by : Enas Khaled, Rwana Ahmed,Hemmat Hamdi,

Designed for : ML project

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Key Activities

- Predict possib -le failures
- Monitor mach- ine performance

Key Resources

- Machine failure datasets
- Python & ML tools
- Historical mainten ance data

Key Partners

- Manufacturing companies
- Maintenance service providers
- Industrial technicians
- Data/AI technol -ogy providers

Value Proposition

- Predicting failures before they happen
- Reduce downtime and high costs
- Help compan- ies schedule techniecans in advance

Customer Segments

- Factories and manufacturi- ng companies
- Production managers
- Industrial companies

Customer Relationships

- Automated failure alerts
- Performance monitoring
- Customer support

Channels

- Web-based platform
- Social media & digital marketing
- AI/technology events

Cost Structure

- Software development
- ML model development
- Data collection and processing
- System maintenance

Revenue Streams

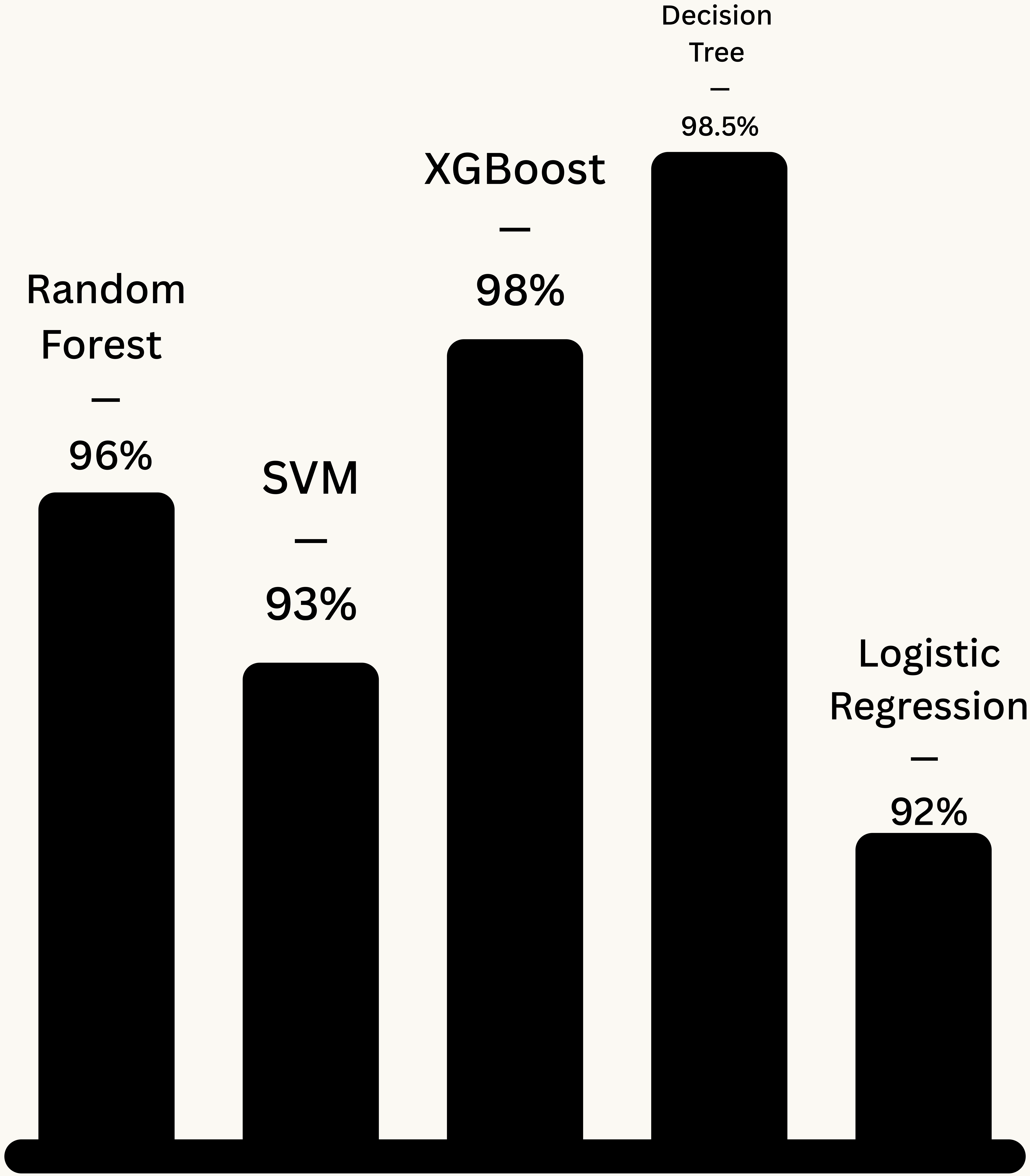
- Monthly/annual subscription
- Premium predictive maintenance plans
- Company licensing
- Customized ML solutions
- Maintenance analytics reports

We tested multiple Machine Learning models to find the one with the highest predictive performance

Models we tested it :

- Logistic Regression
- XGboost
- Random Forest
- SVM
- Decision Tree

XGBoost achieved the highest accuracy among the tested models, But Decision Tree was better in both accuracy and F1_score


Random Forest — 96%
SVM — 93%
XGBoost — 98%
Decision Tree — 98.5%
Logistic Regression — 92%

• Our Team

A team working together to turn machine data into smarter maintenance decisions

Enas Khaled : presintation, EDA, preprocessing, cleaning

Rawan Ahmed : writed and tested models

Hemmat Hamdi : Visualitations,GUI

THANK YOU !